

Overview

This page reviews practical methods for estimating **bus passenger density / occupancy** and translates them into a realistic TransitSense choice.

The research pattern is clear:

1. **Exact full-bus counting is harder than it sounds** in crowded, occluded, moving vehicles.
2. For hackathon use, **occupancy bands** (low / medium / high or seats available / standing room / crowded) are often more realistic and more useful than exact counts.
3. **On-device video processing** is the strongest privacy-friendly path when raw images are not stored or transmitted.
4. **Doorway event counting** is often simpler than whole-cabin density estimation.
5. Stop-level crowd counting and in-bus occupancy are related but distinct problems; TransitSense can benefit from both, but the first MVP should not overpromise exactness.

Core sources reviewed

1. A Privacy-Preserving Edge Computing Solution for Real-Time Passenger Counting at Bus Stops using Overhead Fisheye Camera
2. Automatic Passenger Counting on the Edge via Unsupervised Clustering
3. BusEdge: Efficient Live Video Analytics for Transit Buses via Edge Computing
4. Passenger Counting in Mass Public Transport Systems using Computer Vision and Deep Learning
5. Privacy-centric AI-based real-time storage-less edge computing approaches for passenger counting and action classification on public transport vehicles
6. Video-based automatic people counting for public transport: On-bus versus off-bus deployment
7. Real-Time Passenger Detection and Counting Using Edge AI Camera

Note: Some sources were only accessible through abstracts, repositories, or summary pages rather than fully open detailed PDFs. Where exact metrics were not visible, the method summary below stays cautious.

What the literature consistently says

1. There are three distinct problem styles

A. Doorway counting

Count **boarding and alighting events** and maintain a running occupancy estimate.

B. Cabin density estimation

Estimate how crowded the bus is from **inside-bus imagery**, often using object detection, tracking, or density maps.

C. Stop/platform crowd estimation

Estimate how many people are waiting at a stop or kiosk area using **overhead or wide-angle cameras**.

For TransitSense, these should not be mixed up. The most practical bus-MVP question is:

How crowded is this bus right now, in a privacy-friendly way?

2. Exact counts are fragile; occupancy classes are robust

Papers on deep crowd counting show strong promise, but real deployments still struggle with:

- occlusion,
- motion blur,
- lighting changes,
- door congestion,
- camera angle issues,
- and calibration drift.

That is why an SIH MVP should usually present:

- **Low occupancy**
- **Moderate occupancy**
- **High occupancy**
- **Very crowded**

rather than pretending exact live counts are always trustworthy.

3. Privacy is mainly an architecture choice

The strongest privacy-preserving pattern across sources is:

1. process video **on the edge device**,
2. do **not** store raw video,
3. transmit only aggregate outputs,
4. expose only density level / count / confidence.

The ETH storage-less edge paper and edge-camera sources strongly reinforce that privacy can be designed into the pipeline rather than added later.

Method families

A. Doorway camera counting with line crossing / tracking

Problem

Estimate occupancy by counting each person who boards and alights.

Data

- single camera near bus door
- video stream or frame samples
- running count state

Method

- Detect person entering or leaving.
- Track movement across a virtual line or zone.
- Increase/decrease current occupancy estimate.
- Convert count into occupancy bands if needed.

The edge AI camera source using in-sensor inference plus tracking logic is a strong example of this deployment style.

Results in practice

This is usually easier to operationalize than full-cabin counting because the camera only needs to watch a narrow interaction zone.

Constraints

- Misses happen during heavy crowding or group boarding.
- Errors accumulate over time if correction is not applied.
- Needs a reset/recalibration strategy.

Privacy stance

Strong if raw video is not stored and only counts are retained.

Hardware cost level

Low to medium. A single camera plus Raspberry Pi / edge device can be enough.

Connectivity assumptions

Good for poor network. Counting can run entirely on-bus and only upload aggregates.

Fit for TransitSense

Best practical MVP choice if the goal is live bus occupancy.

B. In-bus occupancy classification from cabin imagery

Problem

Estimate how crowded the bus is from a wider interior view.

Data

- one or more in-cabin camera views
- labelled frames or occupancy classes
- optional head / person annotations

Method

Use lightweight computer vision to classify the scene into density bands, or apply a density-map / crowd-counting model to estimate occupancy.

The TransMilenio deep-learning paper shows the power of density-based crowd counting, while BusEdge demonstrates how bus video analytics can be structured around edge processing and limited bandwidth.

Results in practice

Potentially more informative than doorway counting because it captures overall crowding, not just entry/exit

events.

Constraints

- Occlusion is the main challenge.
- Wide interior coverage may need better camera placement.
- Training data from the actual environment matters a lot.
- Exact counts are less reliable than coarse occupancy bands in student prototypes.

Privacy stance

Medium to strong depending on design. Strong only when inference is on-device and images are not stored or transmitted.

Hardware cost level

Medium.

Connectivity assumptions

Moderate. Best if inference is local and only occupancy labels are uploaded.

Fit for TransitSense

Strong innovation layer. Very good for the pitch, but use occupancy classes instead of claiming precise counts.

C. Overhead stop-level crowd counting

Problem

Estimate crowd size at a stop or kiosk area.

Data

- overhead fisheye or wide-angle view of the waiting area
- edge device for local inference

Method

The Concordia bus-stop study uses an overhead fisheye camera with edge deployment on Jetson hardware to count waiting passengers while preserving privacy.

Results in practice

This is a strong method for **stop crowd estimation**, especially where a kiosk display is part of the experience.

Constraints

- Better for stop crowd than in-bus occupancy.
- Requires mounting and stable field of view.
- Jetson-class hardware may be heavier than a minimal hackathon build.

Privacy stance

Strong if processed locally and only counts are exposed.

Hardware cost level

Medium.

Connectivity assumptions

Good for intermittent network because counts can be generated locally.

Fit for TransitSense

Excellent for future kiosk/stop extensions and possibly demo simulation, but not the easiest first choice for bus-

inside density.

D. Wi-Fi / device-signal counting

Problem

Estimate people count from smartphones or other devices.

Data

- Wi-Fi probe requests
- device-signal observations

Method

The MDPI Sensors paper uses edge clustering on Wi-Fi probe signals.

Results in practice

Interesting as a low-visual-intrusion method, but highly dependent on whether passengers have discoverable devices and whether those signals can be interpreted robustly.

Constraints

- Device ownership $\neq$ person count.
- MAC randomization complicates reliability.
- Stationary nearby devices can create noise.
- Stronger privacy concerns than aggregate CV outputs.

Privacy stance

Weaker than on-device aggregate video methods because device-signal collection can feel more invasive and may raise user-trust issues.

Hardware cost level

Low.

Connectivity assumptions

Good offline.

Fit for TransitSense

Not recommended as the main MVP method. At best, it is a research curiosity or backup signal.

E. Storage-less pose / geometry-based edge analytics

Problem

Infer passenger presence and actions without storing images.

Data

- live camera frames
- pose/joint or geometry features

Method

The ETH privacy-centric paper shows a storage-less edge approach where only transient features are processed for passenger monitoring and action understanding.

Results in practice

This is appealing because it treats privacy as a first-class constraint.

Constraints

- More complex than simple doorway counting.
- Pose quality can degrade in crowded conditions.
- Best for behavior/action understanding rather than only occupancy.

Privacy stance

Very strong.

Hardware cost level

Medium.

Connectivity assumptions

Good, because it is designed around on-device processing.

Fit for TransitSense

Useful as a **privacy design principle** and possible stretch idea, but not the easiest core implementation path.

Comparison table

Design principles for TransitSense

Recommended baseline

Use **doorway-based passenger event counting** to estimate occupancy, then expose only an **occupancy band**.

Why this beats exact count-first thinking:

1. It is easier to build.
2. It works with a single camera.
3. It is easier to explain and debug.
4. It aligns with privacy-first edge processing.
5. The final user output (crowded or not crowded) is what passengers actually care about most.

Recommended advanced option

Add an **in-bus occupancy classification model** that periodically verifies or corrects the doorway count using wide-view snapshots processed locally.

This gives TransitSense a strong hybrid story:

- event counting for continuity,
- scene-based occupancy class for correction,
- aggregate output only.

What to show users

Prefer:

- seats available
- moderate crowd
- standing room
- very crowded

Avoid promising:

- exact passenger count,
- coach-level precision,
- or continuous perfect measurement under all conditions.

Constraints and risks to make explicit

1. **Camera placement matters** more than model choice in many cases.
2. **Crowding at doors causes double counts and missed counts.**
3. **Exact count claims are risky** without extensive calibration.
4. **Edge compute choice matters**; Jetson-class hardware is powerful but may be overkill for a student MVP.
5. **Trust depends on privacy design**; never store identifiable footage.

TransitSense recommendation

Default choice

Build the MVP around a **single doorway camera** with on-device boarding/alighting counting and expose an **occupancy band**.

Add if time permits

Add **in-bus occupancy classification** as a correction or demo comparison layer.

Extension path

Use **overhead stop crowd counting** for kiosk scenarios.

Avoid for first MVP

Do not make **Wi-Fi/device-signal counting** the main strategy.

This choice best matches SIH constraints: privacy-friendly, low-cost, explainable, and robust enough for a believable demo.